

Lesson Overview: Data Security

In an era where data breaches and cybersecurity threats are increasingly common, **securing sensitive information** has never been more crucial. Understanding **data security principles**, from safeguarding data in various states to implementing comprehensive defence strategies, is key to preventing **unauthorised access** and ensuring **data integrity, privacy and security** measures are effectively integrated into data systems.

In this lesson, we'll delve into the **fundamentals of data security**, understanding why it's paramount in preventing unauthorised access and breaches. Through detailed examination of defence-in-depth strategies and data security techniques, this lesson equips us with the knowledge to **implement robust security measures** across various data states and understand the overarching importance of data privacy in security practices.



What You'll Learn:

Data security and its core principles

Learn why data security is critical and explore the fundamental principles that form its foundation.

Defence-in-depth strategies

Discover multi-layered security approaches and how they enhance the protection of data.

Safeguarding data at rest, transit, and during processing

Examine methods to protect data throughout its lifecycle—whether stored, transmitted, or processed.

Data privacy

Understand how data privacy considerations impact the design and implementation of security strategies.

Security practices for Big Data

Learn specialised security practices for big data environments and how to implement layered defences in applications.

Ready to dive in?

Reflect:

Think about a situation where you had to handle sensitive data. What steps did you take to ensure its security? How can the principles you'll learn in this lesson enhance your approach to data protection?